Simulation Assignment 1: Quality Assurance using BESTEST

7B2M0 Building Performance and Energy Systems Simulation

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1 Introduction

Building Performance Simulation (BPS) tools are widely used to predict the energy and thermal performance of buildings. However, the reliability of these predictions depends on the correct implementation of the building model and the ability of the simulation software to accurately represent the relevant physical processes. Therefore, validation and quality assurance are important steps when using building simulation tools.

The objective of this assignment is to evaluate the performance of DesignBuilder using the International Energy Agency (IEA) Building Energy Simulation Test (BESTEST) methodology. The BESTEST Case 600 building is modelled in DesignBuilder according to a predefined set of geometrical, construction, internal gain, infiltration, and HVAC specifications. An annual simulation is then performed to determine the annual heating and cooling energy demands and the peak heating and cooling loads.

The obtained simulation results are compared with the reference results provided by BESTEST. Differences between the DesignBuilder results and the reference values are analysed to identify possible modelling errors, software assumptions, or input settings that may influence the results. Through this process, the assignment aims to demonstrate the importance of validation and troubleshooting in producing reliable building performance simulations.

2 Model Description

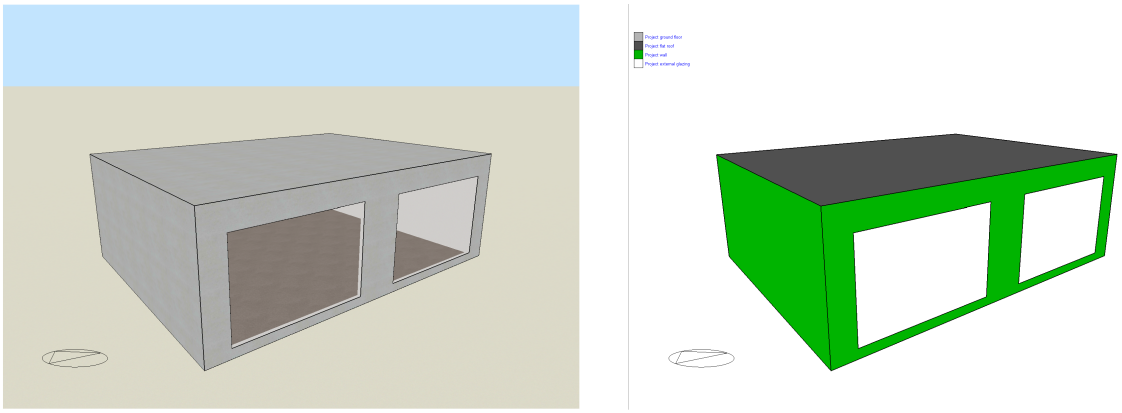
2.1 Model Definition and Geometry

The BESTEST Case 600 building was modelled in DesignBuilder according to the provided reference specifications. The model was defined using the hierarchy

$$\texttt{Building 1} \rightarrow \texttt{Block 1} \rightarrow \texttt{Zone 1},$$

and therefore consists of a single rectangular thermal zone with a floor area of 48 m^2. The ground floor and flat roof both have an area of 48 m^2. The four external walls have gross areas of 21.6, 16.2, 21.6, and 16.2 m^2 at orientations of 0°, 90°, 180°, and 270°, respectively. Two 6 m^2 windows were placed on the 180° facade, resulting in a total glazed area of 12 m^2.

The geometry and surface assignments were checked before defining the thermal properties. Figure 1 shows the resulting model in rendered and construction views.



(a) Rendered view **(b)** Construction view

Figure 1: DesignBuilder model of the BESTEST Case 600 building.

2.2 Construction Specificationss

Custom material and construction templates were created for the external walls, ground floor, and roof because the required BESTEST constructions were not directly available from the standard DesignBuilder templates. For each material, the thermal conductivity λ, density ρ, specific heat capacity c_p, and thickness were entered according to the Case 600 specifications. The resulting layer definitions are summarized in Table 1.

Table 1: Material properties of the opaque envelope constructions.

Element	Material	Thickness (m)	λ (W/mK)	ρ (kg/m^3)	c_p (J/kgK)
Wall	Wood siding	0.0090	0.14	530	900
	Fiberglass quilt	0.0660	0.04	12	840
	Plasterboard	0.0120	0.16	950	840
Floor	Timber flooring	0.0250	0.14	650	1200
	Insulation	1.0030	0.04	12	840
Roof	Plasterboard	0.0100	0.16	950	840
	Fiberglass quilt	0.1118	0.04	12	840
	Roof deck	0.0190	0.14	530	900

The wall was constructed from wood siding, fiberglass insulation, and plasterboard from outside to inside. The ground floor consists of timber flooring above a 1.003 m insulation layer. This unusually thick insulation layer is part of the BESTEST specification and is used to largely decouple the floor thermally from the ground. The roof consists of plasterboard, fiberglass insulation, and roof deck from inside to outside. The implemented cross-sections are shown together in Figure 2.



(a) External wall **(b)** Ground floor **(c)** Flat roof

Figure 2: Opaque construction layers implemented in DesignBuilder.

For the opaque constructions, the thermal absorptance (emissivity) and solar absorptance were set to 0.9 and 0.6, respectively. The prescribed BESTEST surface heat-transfer coefficients were also applied instead of the default DesignBuilder calculation, with

$$h_{\text{in}} = 8.29 \text{ W/m}^2\text{K}, \qquad h_{\text{out}} = 29.3 \text{ W/m}^2\text{K}.$$

After defining the materials and surface properties, the calculated DesignBuilder U-values were checked as a verification of the construction inputs. The resulting values were approximately 0.510 W/m^{2}K for the walls, 0.039 W/m^{2}K for the ground floor, and 0.319 W/m^{2}K for the roof. The wall construction was assigned to all four external facades, while the floor and roof constructions were assigned to their respective 48 m^2 surfaces.

2.3 Glazing

The two windows were modelled using the Case 600 double-glazing specification. Each window has an area of 6 m^2 and consists of two 3.175 mm clear glass panes separated by a 13 mm air gap. No curtains, blinds, or other shading devices were included.

The main glazing inputs and calculated properties are summarized in Table 2. The glass panes were defined using DesignBuilder's broadband optical model.

Table 2: Main properties of the glazing system.

Property	Value
Number of panes	2
Glass thickness	3.175 mm
Air-gap thickness	13 mm
Glass conductivity	1.06 W/(mK)
Solar transmittance (single pane)	0.86156
Solar reflectance (inside/outside)	0.05844
Visible transmittance (single pane)	0.898
Long-wave emissivity	0.90
Calculated glazing U-value	2.876 W/m^2K
Calculated SHGC	0.799

The calculated U-value and SHGC differ slightly from the BESTEST target values of approximately 3.00 W/m^2K and 0.787, respectively. These differences are considered later when discussing deviations between the simulation and the BESTEST reference results.

2.4 Internal Conditions and Infiltration

The zone was defined as unoccupied, so no occupant heat gains were included. The prescribed constant internal sensible gain was represented using office equipment with a power density of 4.39 W/m^2 and an `On 24/7` schedule. The radiant fraction was set to 0.60, corresponding to a 60% radiative and 40% convective distribution, with no latent component.

A constant infiltration rate of 0.5 ACH was applied throughout the simulation. Mechanical and natural ventilation were disabled so that the specified infiltration represents the outdoor air exchange.

The heating and cooling setpoints were fixed at

$$T_{\text{heat}} = 20^\circ\text{C}, \qquad T_{\text{cool}} = 27^\circ\text{C}.$$

The setback temperatures were set equal to these values, resulting in constant temperature limits throughout the year.

2.5 HVAC System

A custom `HVAC - BESTEST` template was created to represent the ideal heating and cooling system specified for Case 600. Heating and cooling were available continuously using an `On 24/7` schedule, with no capacity limitation. The system therefore supplies the thermal load required to prevent the zone temperature from falling below 20°C or rising above 27°C.

Heating was defined as 100% convective, and both the heating and cooling seasonal CoP were set to 1.0. Natural gas was selected as the heating fuel and grid electricity for cooling. Mechanical ventilation, natural ventilation, humidification, and dehumidification were disabled. This simplified configuration was used so that the reported heating and cooling loads mainly represent the thermal response of the Case 600 building rather than the performance of a detailed HVAC plant.

2.6 Simulation Setup

After defining the Case 600 model, an annual EnergyPlus simulation was performed through DesignBuilder. The simulation period was set from 1 January to 31 December using the provided weather data. Heating and cooling were available continuously throughout the simulation according to the 20°C heating and 27°C cooling setpoints defined in the model.

Both run-period and hourly results were exported from DesignBuilder. The run-period results were used to obtain the total annual sensible heating and cooling energy demands, while the hourly results were used to determine the maximum heating and cooling loads. This distinction is important since the annual results represent accumulated energy use in kWh, whereas the hourly results allow the maximum instantaneous hourly load in kW to be identified.

The annual heating and cooling demands were converted from kWh to MWh. Since DesignBuilder reports sensible cooling with a negative sign, the absolute value was used when reporting the cooling energy demand and peak cooling load.

3 Results

3.1 Simulation Results

The main results obtained from the annual simulation are summarized in Table 3. The annual sensible heating demand was 4.572 MWh, while the annual sensible cooling demand was 8.961 MWh. The maximum hourly heating and cooling loads used for the BESTEST comparison were 4.014 kW and 7.813 kW, respectively.

Table 3: Main simulation results for BESTEST Case 600.

Output	Result	Unit
Annual heating demand	4.572	MWh
Annual cooling demand	8.961	MWh
Peak heating load	4.014	kW
Peak cooling load	7.813	kW

Figure 3 shows the monthly variation of the main thermal variables and heat-balance components throughout the simulation year. A clear seasonal pattern can be observed. Heating demand is highest during the colder months and decreases towards the middle of the year. Cooling remains present throughout the year but becomes more dominant during the warmer period.

The monthly results also show the contribution of solar gains through the exterior windows. These gains decrease towards the middle of the year and increase again during the later months. This behaviour is related to the orientation of the glazing and the seasonal change in solar position. Together with the internal gains, the solar gains contribute to the heat that must be removed by the cooling system.

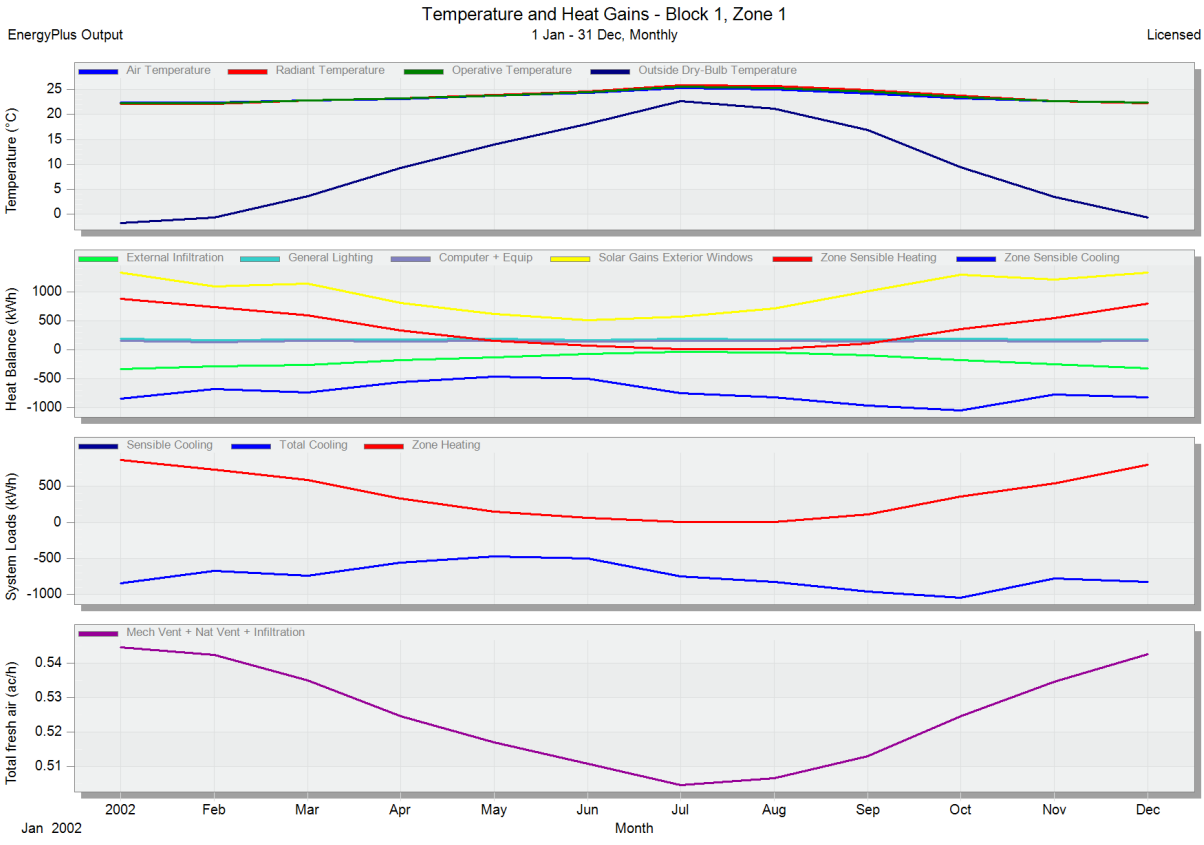


Figure 3: Monthly temperature, heat-balance, system-load, and fresh-air results for BESTEST Case 600.

The annual heat-balance results provide some additional information about the sources of heat entering the zone. Approximately 10.84 MWh of heat addition was associated with the exterior windows over the simulation year. Internal heat gains also contributed to the zone heat balance, including approximately 1.75 MWh from equipment and 2.14 MWh from lighting. These components are particularly

relevant when analysing the cooling results, since the resulting heat gains have to be removed from the zone to maintain the cooling setpoint.

3.2 Comparison with BESTEST Reference Results

The simulation results were compared with the minimum and maximum reference values provided for BESTEST Case 600. Table 4 presents the DesignBuilder results together with the corresponding BESTEST reference ranges. For results outside the reference range, the absolute difference from the nearest acceptable limit is also reported.

Table 4: Comparison of DesignBuilder results with the BESTEST Case 600 reference ranges.

Output	DB Result	BESTEST Min.	BESTEST Max.	Difference	Status
Annual heating demand (MWh)	4.572	4.296	5.709	–	Within range
Annual cooling demand (MWh)	8.961	6.137	7.964	+0.997	Above range
Peak heating load (kW)	4.014	3.437	4.354	–	Within range
Peak cooling load (kW)	7.813	5.965	6.812	+1.001	Above range

The heating results fall within the BESTEST reference limits. The annual heating demand of 4.572 MWh lies within the reference range of 4.296–5.709 MWh. Similarly, the peak heating load of 4.014 kW is within the reference range of 3.437–4.354 kW.

A different result is obtained for cooling. The annual cooling demand of 8.961 MWh exceeds the upper BESTEST reference value of 7.964 MWh by 0.997 MWh, corresponding to approximately 12.5% above the upper limit. The peak cooling load of 7.813 kW also exceeds the upper reference value of 6.812 kW by 1.001 kW, or approximately 14.7%.

Overall, two of the four investigated outputs fall within the BESTEST reference ranges. Both heating-related results satisfy the reference limits, while both cooling-related results are above their respective upper limits. The fact that the discrepancy occurs for both annual cooling demand and peak cooling load indicates that the cooling behaviour of the model should be investigated further rather than treating the difference as only an isolated peak event. Possible input errors, modelling assumptions, and DesignBuilder defaults that may have contributed to this discrepancy are therefore examined in the following troubleshooting section.

4 Troubleshooting Log

The initial simulation showed good agreement with the BESTEST reference range for heating, but both cooling-related results were above the acceptable limits. The annual cooling demand was initially 8.960 MWh, exceeding the upper reference limit of 7.964 MWh by 0.996 MWh (12.5%). Similarly, the peak cooling load was 7.813 kW, which was 1.001 kW (14.7%) above the upper limit of 6.812 kW.

Since both discrepancies were related to cooling, the model inputs contributing to internal and solar heat gains were investigated. The geometry, infiltration rate, temperature setpoints, HVAC operation, construction properties, and simulation settings were checked against the Case 600 specifications. During this process, an unintended lighting load inherited from the DesignBuilder office template was identified. BESTEST Case 600 specifies a constant internal sensible gain of 200 W, whereas the model contained both the specified equipment gain and an additional lighting load. This additional internal heat gain increased the cooling requirement of the zone.

The lighting contribution was therefore removed while retaining the equipment load representing the required 200 W internal gain. The model was then simulated again for the complete annual period using the same remaining model and simulation settings.

Table 5 compares the results before and after this correction.

Table 5: Simulation results before and after troubleshooting.

Output	Before	After	BESTEST Range	Final Status
Annual heating demand (MWh)	4.572	4.936	4.296–5.709	Within range
Annual cooling demand (MWh)	8.960	7.462	6.137–7.964	Within range
Peak heating load (kW)	4.014	4.015	3.437–4.354	Within range
Peak cooling load (kW)	7.813	7.072	5.965–6.812	Above range

The correction had its largest effect on the cooling results. Annual cooling demand decreased from 8.960 to 7.462 MWh, a reduction of approximately 16.7%, bringing it within the BESTEST reference

range. Peak cooling load also decreased from 7.813 to 7.072 kW. Its deviation from the upper reference limit was reduced from 1.001 to 0.260 kW, corresponding to approximately 3.8% above the upper limit. The heating results remained within the reference range before and after the correction.

The troubleshooting process therefore identified the unintended lighting load as an important source of the original cooling discrepancy. Removing this additional internal gain substantially improved agreement with the BESTEST results. Three of the four evaluated outputs are within the reference ranges after correction, while the remaining peak cooling discrepancy is relatively small and may result from differences in software modelling assumptions or other simulation defaults.

5 Conclusion

This assignment evaluated how well DesignBuilder can reproduce the expected results for BESTEST Case 600. After troubleshooting the model, three of the four investigated outputs were within the BESTEST reference ranges. The final annual heating and cooling demands were 4.936 MWh and 7.462 MWh, respectively, and the peak heating load was 4.015 kW. The peak cooling load remained slightly above the reference range at 7.072 kW, which is approximately 3.8% above the upper limit of 6.812 kW.

The troubleshooting process showed that relatively small modelling mistakes can have a significant effect on the simulation results. In the initial model, an additional lighting load was inherited from the DesignBuilder office template, even though the required 200 W internal gain had already been included through the equipment load. After removing this lighting load, the annual cooling demand decreased from 8.960 to 7.462 MWh and moved inside the BESTEST range. The peak cooling load also decreased from 7.813 to 7.072 kW. This shows that internal gains can have a strong effect on cooling predictions and that default template settings need to be checked carefully rather than assumed to be correct.

A small difference still remained for the peak cooling load. One possible contributor is that some properties could not be reproduced exactly in DesignBuilder. For example, the implemented glazing had a calculated U-value of 2.876 W/m^2K and an SHGC of 0.799, compared with the BESTEST target values of approximately 3.00 W/m^2K and 0.787. In addition, differences in how DesignBuilder and EnergyPlus handle heat transfer, solar gains, and other simulation calculations may affect peak values. Therefore, the remaining 0.260 kW difference cannot be linked to one specific input based on the current analysis.

Overall, DesignBuilder was found to be suitable for modelling the BESTEST Case 600 building, as the final results showed good agreement with the reference ranges. However, the exercise also showed that obtaining reasonable results does not automatically mean that the model has been set up correctly. Default templates and inherited settings can introduce inputs that are easy to overlook. Therefore, checking individual model inputs and comparing simulation outputs against reference cases such as BESTEST are important steps for obtaining reliable building performance simulation results.

References

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